BIOMARKER DISCOVERY AND VOICE-BASED CLASSIFICATION OF PARKINSON'S DISEASE: A COMPARATIVE MACHINE LEARNING AND HYPERPARAMETER OPTIMIZATION STUDY

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ABSTRACT

Early identification of Parkinson's disease using non-invasive voice measurements requires models capable of representing subtle, potentially non-linear patterns in speech. This project analyzes 195 biomedical voice recordings and compares Logistic Regression, a Linear Support Vector Machine (SVM), and an optimized Radial Basis Function (RBF) kernel SVM. Logistic Regression achieved 82.05% test accuracy, while the Linear SVM achieved 87.18%. A five-fold grid search selected an RBF SVM with C = 100 and gamma = 0.1, yielding 92.31% test accuracy. On the held-out test set, the model achieved 100.00% specificity and 100.00% precision, with 90.32% sensitivity. These results indicate promising classification performance on this dataset; however, the small test sample and the absence of external validation mean that the model should not be interpreted as a clinically validated diagnostic system.

1. PROJECT OVERVIEW AND RATIONALE

Voice impairment (dysphonia) can be associated with Parkinson's disease. Changes in pitch, frequency fluctuations (jitter), and amplitude instability (shimmer) may provide measurable signals for classification. The project addresses two questions: whether vocal biomarker patterns can be separated by linear decision boundaries or require non-linear combinations, and how hyperparameter choices such as C and gamma affect false-positive and false-negative outcomes.

2. DATASET AND EXPLORATORY ANALYSIS

The analysis uses the UCI Parkinson's dataset, containing 195 voice recordings, 22 continuous vocal attributes, a participant identifier (name), and a binary diagnosis label (status: 1 = Parkinson's disease, 0 = healthy).

- **Observations:** 195 total; 147 Parkinson's-positive (75.38%) and 48 healthy (24.62%).
- **Missing values:** None reported across the 24 columns.
- **Feature types:** 22 continuous numerical features, one binary target, and one string identifier.

Vocal feature groups

- **Pitch and frequency:** MDVP:Fo(Hz), MDVP:Fhi(Hz), MDVP:Flo(Hz).
- **Frequency variation (jitter):** MDVP:Jitter(%), MDVP:Jitter(Abs), MDVP:RAP, MDVP:PPQ, Jitter:DDP.
- **Amplitude variation (shimmer):** MDVP:Shimmer, MDVP:Shimmer(dB), Shimmer:APQ3, Shimmer:APQ5, MDVP:APQ, Shimmer:DDA.
- **Noise ratios:** NHR and HNR.
- **Non-linear voice dynamics:** RPDE, DFA, spread1, spread2, D2, PPE.

Comparison of class averages showed that healthy participants had a higher mean fundamental frequency (Fo) of approximately 181.94 Hz, compared with 145.18 Hz among participants with Parkinson's disease. Mean MDVP:Jitter(%) was 0.003866 in healthy participants and 0.006989 in the Parkinson's group. Mean HNR was 24.63 dB for healthy participants and 21.82 dB for the Parkinson's group.

3. PREPROCESSING PIPELINE

To reduce the risk of data leakage, the string identifier (name) was removed and the 22 continuous vocal features were separated from the target variable (status). The data were split into an 80% training partition (n = 156) and a 20% held-out test partition (n = 39), using random_state = 2. StandardScaler was fitted only on the training partition; the learned parameters were then used to transform both partitions.

$$X_scaled = (X - \mu_train) / \sigma_train$$

4. MODELING EXPERIMENTS AND FINDINGS

4.1 Logistic Regression

Logistic Regression was used as the linear benchmark on the scaled features.

- Training accuracy: 87.18% (136/156).
- Test accuracy: 82.05% (32/39).

4.2 Linear Support Vector Machine

A Linear SVM was evaluated to assess whether a linear decision boundary could better separate the two classes.

- Training accuracy: 88.46% (138/156).
- Test accuracy: 87.18% (34/39).

The Linear SVM test accuracy exceeded Logistic Regression by 5.13 percentage points. The remaining errors motivated evaluation of a non-linear decision boundary.

5. HYPERPARAMETER OPTIMIZATION: RBF KERNEL SVM

An SVM with a Radial Basis Function kernel was used to model non-linear relationships among vocal features. The kernel is defined as:

$$K(x, x') = \exp(-\gamma \|x - x'\|^2)$$

A five-fold cross-validated GridSearchCV evaluated the following parameter values:

- Regularization parameter C: 0.1, 1, 10, 100.
- Kernel scale gamma: 0.001, 0.01, 0.1, 1, 'scale', and 'auto'.

Selected grid-search mean cross-validation accuracies reported for representative combinations are shown below.

C	γ = **0.001**	γ = **0.1**	γ = **1**
0.1	0.7435	0.7435	0.7435
1	0.7435	0.8782	0.7821
10	0.8464	0.9292	0.8075
100	0.8655	0.9357	0.8075

- Selected parameters: C = 100 and gamma = 0.1.
- Optimized model training accuracy: 100.00% (156/156).
- Optimized model test accuracy: 92.31% (36/39).

6. DIAGNOSTIC PERFORMANCE EVALUATION

Accuracy alone does not describe the balance between false positives and false negatives. The following metrics were reported for the held-out test set (n = 39).

Metric	**Result**	**Interpretation**
Accuracy	92.31%	Correct predictions across the test sample.

Metric	Result	Interpretation
Sensitivity (recall)	90.32%	28 of 31 actual Parkinson’s cases detected.
Specificity	100.00%	All 8 healthy controls correctly identified; no false positives in this test set.
Precision (PPV)	100.00%	28 of 28 positive predictions were correct.

Confusion matrix

Actual class	Predicted healthy (0)	Predicted Parkinson’s (1)
Healthy	8	0
Parkinson’s	3	28

The classification report recorded the following scores:

Class	Precision	Recall	F1-score	Support
Healthy (0)	0.73	1.00	0.84	8
Parkinson’s (1)	1.00	0.90	0.95	31
Micro average	0.92	0.92	0.92	39
Macro average	0.86	0.95	0.90	39
Weighted average	0.94	0.92	0.93	39

7. CONCLUSIONS AND FUTURE WORK

- **Non-linear patterns:** The optimized RBF SVM improved test accuracy by 10.26 percentage points over Logistic Regression and 5.13 percentage points over the Linear SVM.
- **Screening metrics:** The test set showed 100.00% specificity and 90.32% sensitivity. Because the test set contained only 39 observations, these estimates are uncertain and require validation on larger, independent samples.
- **Future work:** Evaluate tree-based models such as Random Forest and XGBoost, and use SHAP analysis to investigate the contributions of features such as PPE, spread1, and jitter.

This project demonstrates a comparative machine-learning workflow for voice-based classification on a public dataset. The reported performance should be treated as exploratory and is not evidence of readiness for clinical diagnosis without independent validation.